Brazil–Canada Trade Data Quality Review

Diagnostics & Cleaning Recommendations

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Executive Summary

This report evaluates the quality of Brazil–Canada import/export data to ensure reliable pricing benchmarks, volume tren

Main findings:

- **~43.1% imports & ~63.5% exports have zero quantity → blocks unit price calculation**
- **2 products use multiple units of measure → prevents accurate aggregation**
- **5,023 extreme price records detected → can distort averages**

Fixing these three issues would greatly increase confidence in trade analytics and reporting.

Data Quality Snapshot

Valid records: 63,716 (imports) + 14,964 (exports)

Zero quantity rate: 43.1% / 63.5%

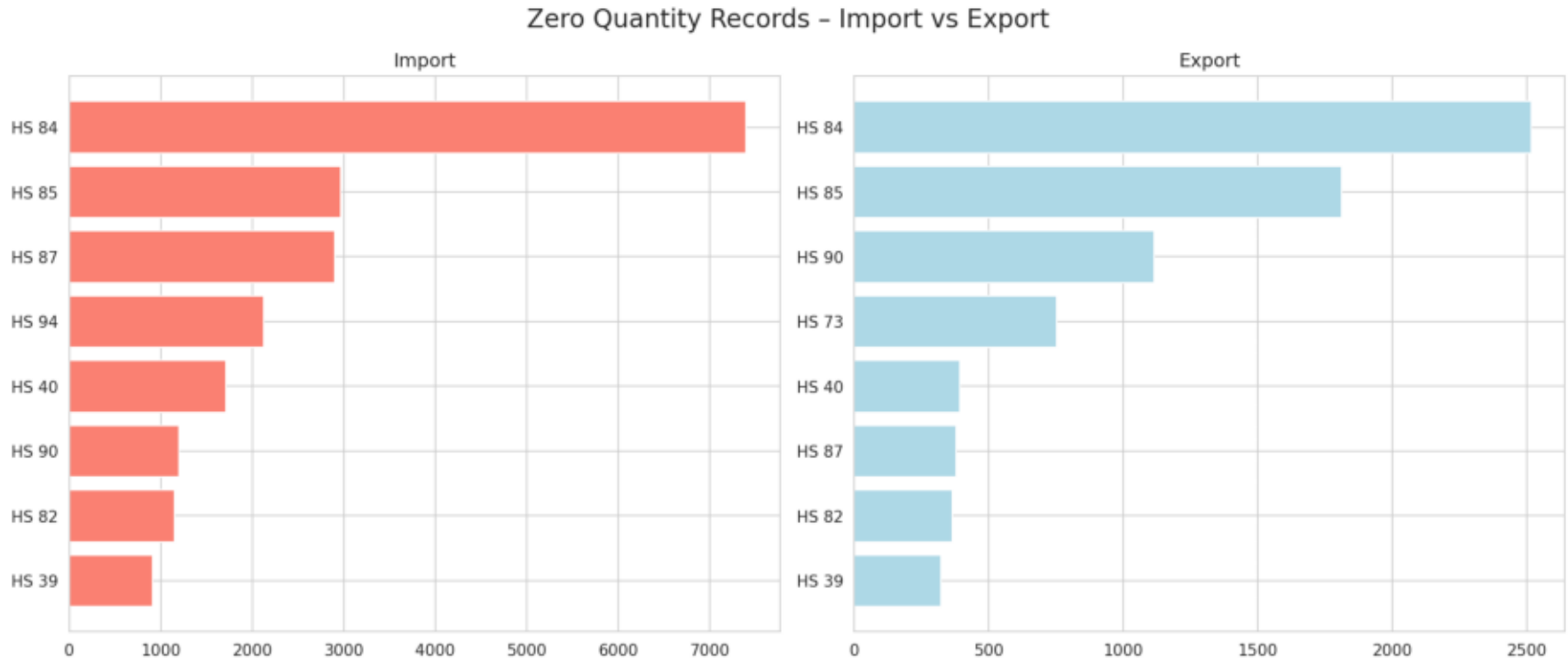
Products with multiple UOMs: 2

Price outliers detected: 5,023

Other notes: Minor date parsing issues and text encoding problems (low impact)

Priority: Zero quantities > UOM consistency > Price outliers

Why are so many records showing zero quantity?



What We See

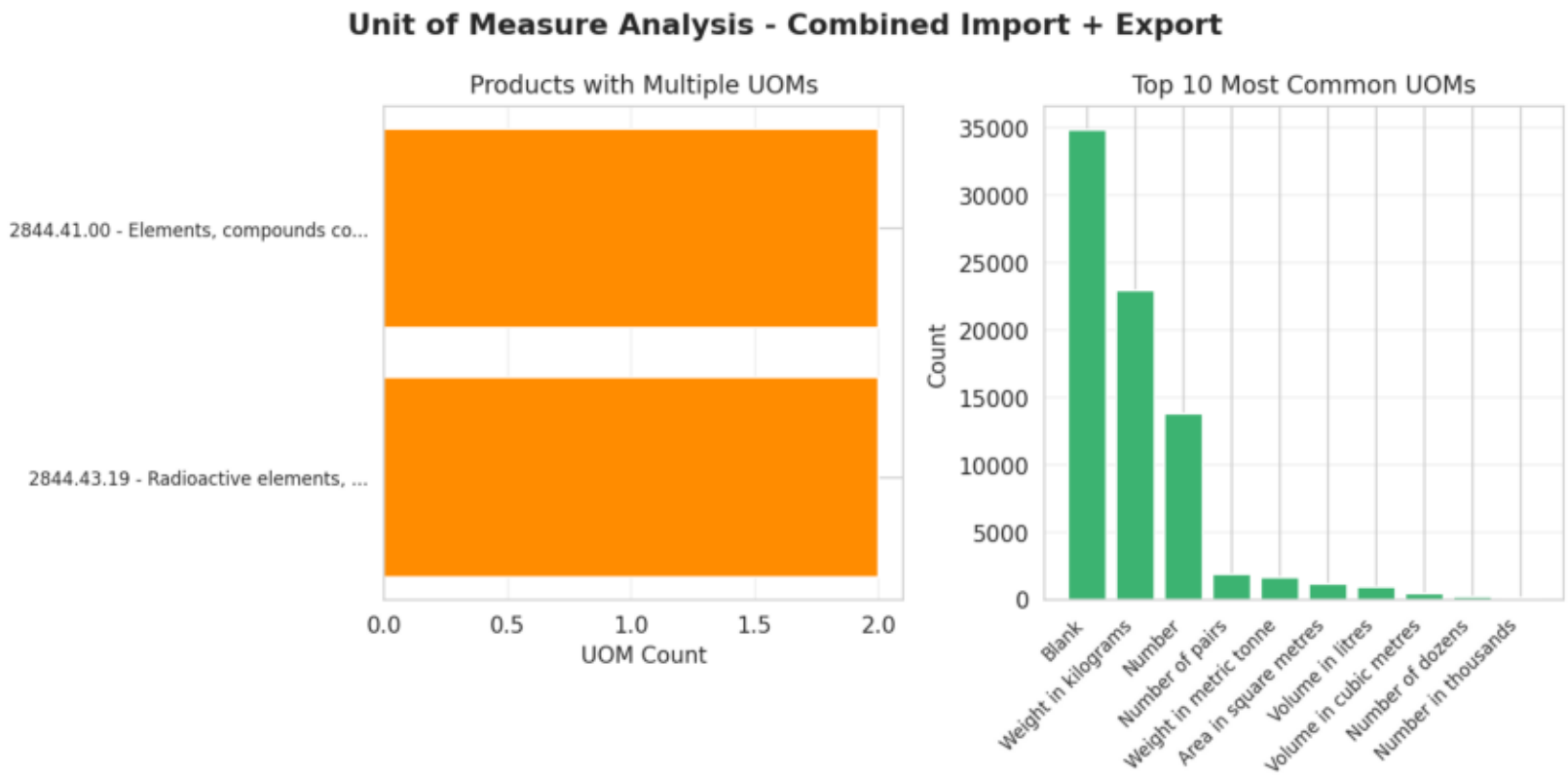
- Affects ~43.1% imports & ~63.5% exports
- Prevents calculation of reliable unit prices
- Concentrated in specific HS sections – targeted cleaning has high ROI
- Persistent across months/years – not a temporary glitch

Recommended Actions

→ Flag zero-quantity records and exclude from price benchmarking

→ For high-value records, impute quantity using HS-section median price

Why can't we accurately sum quantities across products?



What We See

- 2 products use multiple units of measure
- Prevents reliable total volume reporting and comparison
- Weight units (kg, MT) are most common – good starting point for standardization

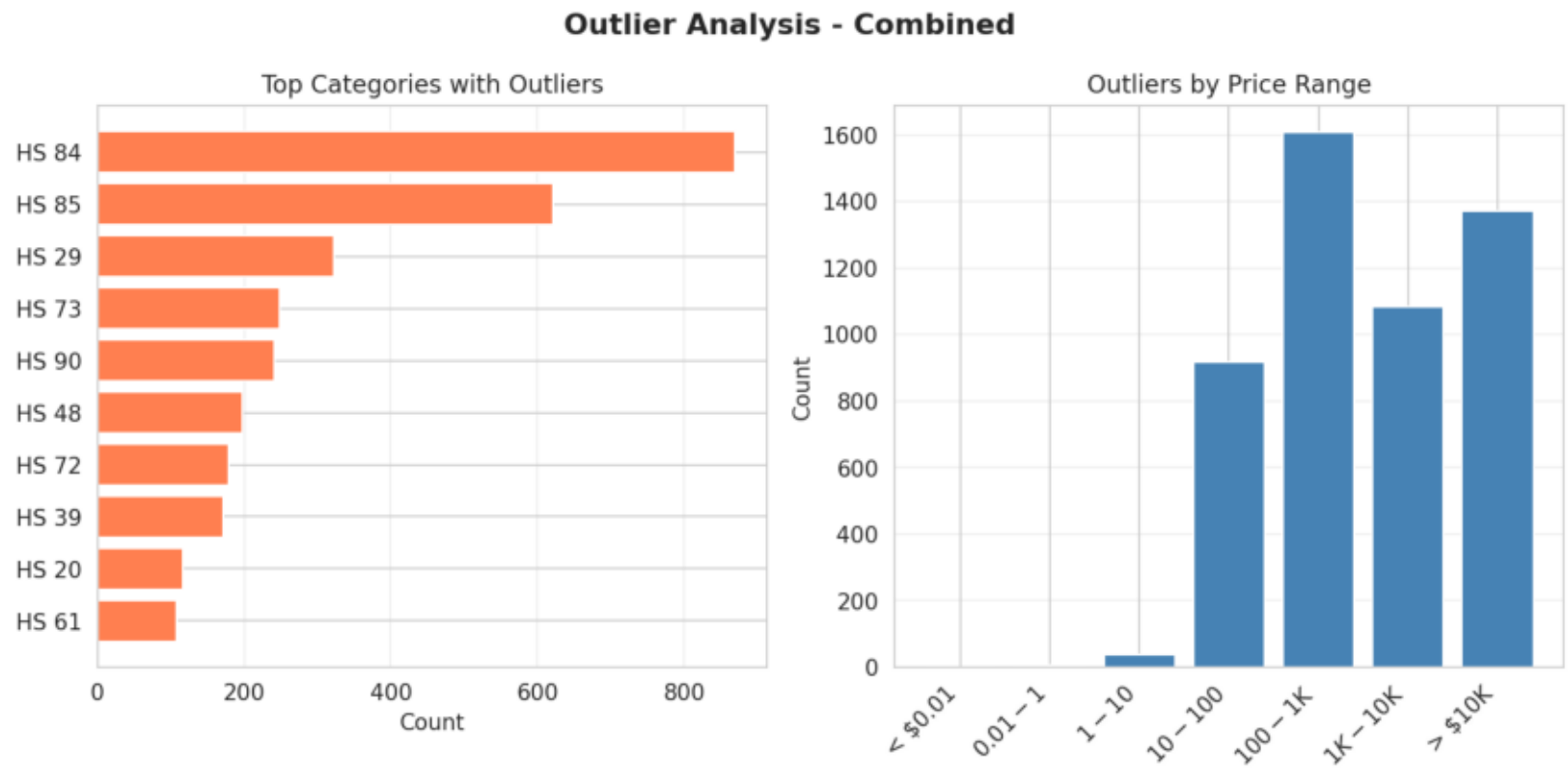
Recommended Actions

→ Define one preferred unit per HS code family

→ Create & apply conversion table (e.g. 1 MT = 1000 kg)

→ Medium effort – high value for volume-based analytics

Why do some unit prices look unrealistically high/low?



What We See

- 5,023 records flagged as price outliers
- Mostly extreme high values – likely errors or rare transactions
- Can heavily skew average prices and benchmarks if not handled

Recommended Actions

→ Flag outliers using HS-section specific IQR rules

→ Winsorize (cap) or exclude from average price calculations

→ Quick win – low risk, immediate improvement in reliability

Other / Lower Priority Issues

- Unparseable dates: ~0 records (fix parsing logic)
- Non-ASCII characters (accents) in descriptions – minor risk unless breaking exports
- No widespread duplicates or completely empty columns
- Core fields (value, commodity, period) mostly usable after basic cleaning

Recommended Priority Action Plan

1. Zero Quantity Handling – Highest impact (exclude or impute selectively)

2. Standardize Units of Measure per HS code + conversion table

3. Flag / Winsorize Price Outliers using category rules

4. Improve date parsing & normalize text (lower priority)

Estimated effort: 2–4 weeks depending on imputation depth

Next Steps & Timeline

Review & approve top 3 actions (this week)

Assign owner and detailed plan (next 1–2 weeks)

Implement cleaning steps in data pipeline (3–4 weeks)

Re-run quality diagnostic to measure improvement

Integrate cleaned data into regular trade reports

Appendix – Detailed Charts

(Skip during presentation – available for reference)

Appendix: Unit Price Distribution (Combined)

What We See

- Wide range – log scale recommended
- Long tail can distort averages

Recommended Actions

→ Use log transformation

→ Consider price segmentation